

Molecular Pathogenesis Assignment

Please pick an inherited disease from the assigned list and answer the following questions.

Disease Name: Alagille Syndrome

Disease Phenotype (be brief and list only the main features):

Alagille Syndrome was originally described as a reduction in vascular ducts, in addition to 5 main clinical abnormalities: cholestasis, cardiac disease, skeletal abnormalities, ocular abnormalities, and a characteristic facial phenotype. A distinct feature of this disorder is a wide spectrum of clinical severity ranging from life-threatening liver or cardiac disease to only subclinical manifestations.

Inheritance Pattern most commonly involved: Autosomal Dominant

Gene Name and Acronym of gene most commonly involved in the disorder: JAG1 gene or rarely NOTCH2

Type of mutation/s found in the gene (e.g. missense, nonsense etc):

Majority of mutations result of loss of function, often caused by a frameshift which resulted in a premature stop codon resulting in truncated protein or nonsense mediated decay

What is the pathogenesis based on genetic information? (i.e. LOF, GOF etc)? LOF

What is the evidence to support the pathogenesis?

Functional studies:

[High et al. \(2008\)](#) found that endothelial-specific deletion of Jag1 in mice resulted in embryonic lethality and cardiovascular defects, recapitulating the Jag1-null phenotype.

Population:

[Spinner et al. \(2001\)](#) summarized data on 233 Alagille syndrome patients reported with mutations in JAG1. The data had been published by 7 different laboratories in Europe, the United States, Australia, and Japan. Mutations were demonstrated in 60 to 75% of patients with a clinically confirmed diagnosis of Alagille syndrome.

Take one mutation and describe what is happening on the molecular level as to how the mutation is affecting the RNA or protein. If none is known, what is a hypothesis for the pathogenic mechanism?

The VOI is c.1656del, it is a single base pair deletion of C nucleotide at base pair 1656 resulting in a frameshift ultimately causing the formation of a premature stop codon and truncated transcript.

If known, describe how mutations in this gene is altering cellular function to cause the specific phenotype of the disorder. If not known, what is the hypothesis as to how the gene is altering cell function in causing the phenotype?

Please list your references here. Though you can refer to websites and review papers, they should always refer to primary references which you should also use here.